Convergence of Policy Gradient Methods to Nash Equilibria in Stochastic Games

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2026

Why Does This Matter?

Algorithmic pricing is everywhere.

Airlines, ride-sharing, e-commerce: prices are set by AI, in real time, millions of times a day.

These algorithms are not isolated — they **interact**. Each one watches the others and adapts. They are, effectively, playing a game.

The concern regulators have:

- Can these algorithms learn to sustain artificially high prices...
- ...*without anyone programming them to collude?*

Answering this rigorously requires understanding whether competitive learning processes **converge to a Nash equilibrium** — and **this is exactly what this dissertation proves**.

The Reinforcement Learning Loop

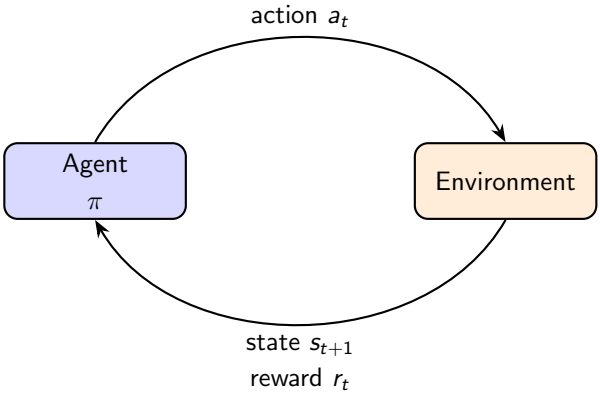
An agent interacts with an environment step by step:

1. Observe current **state** s_t
2. Choose **action** a_t via policy π
3. Receive **reward** $r_t = R(s_t, a_t)$
4. Environment moves to new state

Markov property: the next state depends only on (s_t, a_t), not on the full history.

$$s_{t+1} \sim \mathcal{P}(\,\cdot \mid s_t,\ a_t\,)$$

Goal: find π maximising $V^\pi = \mathbb{E}\big[\sum_t r_t\big]$.



(This is a **Markov Decision Process**. Stochastic games generalise it to N agents.)

What is Known (and What Isn't)

Solved in special cases:

- All players share the same goal (potential games)
- Two players, strictly opposed (zero-sum games, under extra conditions)

Convergence guarantees exist in these cases.

Open — the general case:

- Mixed motives (partly competing, partly cooperating)
- Many players
- The environment has multiple states

No convergence guarantee existed for this setting.

This dissertation: prove convergence in the general case, and make it *global* (independent of where you start).

Stochastic Games: N Agents in a Shared MDP

Replace the single agent with N agents, all learning at once.

$$\mathcal{G} = \langle \underbrace{\mathcal{N}}_{\text{players}}, \underbrace{\mathcal{S}}_{\text{states}}, \underbrace{(\mathcal{A}_i)}_{\text{actions}}, \underbrace{(R_i)}_{\text{rewards}}, \underbrace{\mathcal{P}}_{\text{transitions}}, \underbrace{\rho}_{\text{start}} \rangle$$

$\mathcal{N} = \{1, \ldots, N\}$	players	
$\mathcal{S}$	shared state space	e.g. price levels, board position
$\mathcal{A}_i$	actions for player i	joint action $a = (a_1, \ldots, a_N)$
$R_i(s, a) \in \mathbb{R}$	reward to player i	each player has their *own*
$\mathcal{P}(s' \mid s, a)$	transition kernel	same Markov dynamics as before
ρ	initial state distribution	

Policy $\pi_i(a \mid s)$: probability of action a in state s. **Value:** $V_i(\pi) = \mathbb{E}\left[\sum_t R_i(s_t, a_t)\right]$.
Nash equilibrium π^*: no player gains by deviating unilaterally.
Think: a stable resting point — everyone plays their best response to everyone else.

Policy Gradient: The Update Rule

Each agent does gradient ascent on its own value, projected onto the strategy space:

$$\pi_{n+1} = \mathrm{proj}_{\Pi}\big(\pi_n + \gamma_n \, \hat{v}_n\big)$$

- $\hat{v}_n \approx \nabla_{\pi_i} V_i(\pi_n)$ for each agent i — estimated from sampled trajectories (REINFORCE)
- $\gamma_n \to 0$ — decaying step size

Theorem 3.1 (Local Convergence)

If π_0 starts close enough to a *stable* Nash equilibrium π^* (stable = small deviations self-correct),

$$\mathbb{P}\big(\pi_n \to \pi^* \text{ as } n \to \infty\big) = 1$$

Why It Converges: The Lyapunov Argument

Define the **energy** $E_n = \|\pi_n - \pi^*\|^2$ (squared distance to Nash).

The update rule gives:

$$E_{n+1} \;\leq\; E_n \underbrace{- 2\gamma_n \langle v(\pi_n),\, \pi_n - \pi^* \rangle}_{\text{always negative near } \pi^*} + \underbrace{\text{noise} + \text{bias}}_{\text{controlled}}$$

Energy *tends to decrease* — with controlled random fluctuations.

A classical lemma (Robbins–Siegmund) turns this into almost-sure convergence.

Our contribution: a fully rigorous, self-contained version of this argument — every probability space and random variable verified explicitly.

The Initialisation Problem

Local convergence only kicks in if you *start close enough* to Nash.
In practice, agents have no idea where Nash equilibria are.

Solution: *Random restarts.*

Algorithm: Restart-PG

1. Pick a random starting strategy $\pi_0^{(k)}$ uniformly
2. Run gradient ascent for T steps $\rightarrow$ candidate $\hat{\pi}^{(k)}$
3. If converged: stop. Else: $k \leftarrow k+1$, go to 1.

Key quantity: the *basin probability* p = fraction of the strategy space that flows toward *some* Nash equilibrium under gradient ascent. As long as $p > 0$, restarts eventually succeed.

Main Result: Almost-Sure Convergence

Theorem 4.2 (Original Contribution)

Under mild regularity conditions, Restart-PG converges to a Nash equilibrium **almost surely** (= with probability 1):

$$\mathbb{P}\big(K^* < \infty\big) = 1$$

with geometric tail: $\mathbb{P}(K^* > k) \leq (1-p)^k$

- K^* = number of restarts until convergence; $\mathbb{E}[K^*] \leq 1/p$
- **Equilibrium selection:** running K restarts and keeping the best outcome approaches the **Pareto optimum** — the outcome where no one can be made better off without making someone else worse off

Blessing of Dimensionality

Worry: Basin volume shrinks exponentially in dimension $d = \dim(\Pi)$.

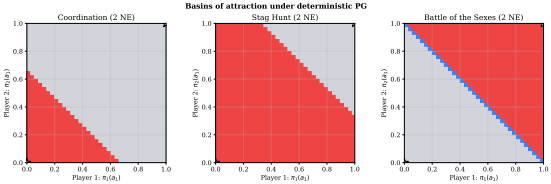
Key insight: When Nash strategies are *sparse* (only k actions matter, $k \ll d$):

- The attraction basins extend broadly in the $d - k$ "irrelevant" directions — they are large in practice
- Empirically observed restart cost: $\mathcal{O}(k \log d)$ instead of $\mathcal{O}(d)$
- Confirmed empirically: a single restart sufficed ($K^* = 1$) for all tested games up to $d = 7$ strategy dimensions

Implication: Restarts are practical, not just theoretical.

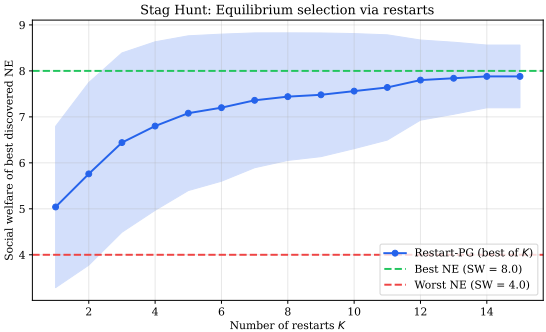
Single-State Games: Basins and Equilibrium Selection

Basins of attraction



Coloured regions = which Nash equilibrium you land on from each starting point. They tile the whole space: a single random restart almost always suffices ($p \approx 1$).

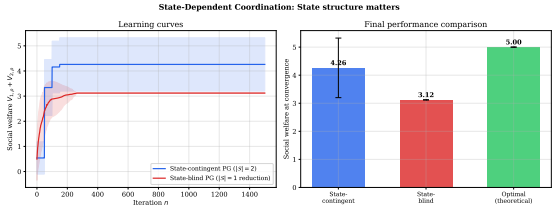
Equilibrium selection



Stag Hunt: 2 hunters. Both cooperate → stag (best). One defects → defector gets rabbit, cooperator gets nothing. Best-of-K restarts reliably finds the cooperative equilibrium.

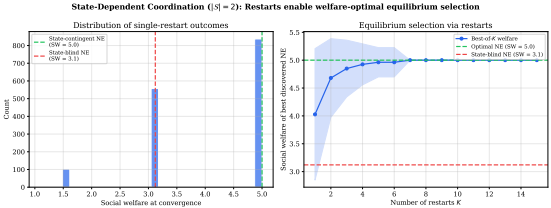
Two-State Game: Key Findings

State structure matters



Agents using state: SW $= 5.0$.
Agents ignoring state: SW ≈ 3.1.
38% welfare gap from collapsing the state space.

Restarts select the best equilibrium



Single restarts land on different Nash equilibria. Best-of-K reliably finds the Pareto-optimal outcome.

Summary of Contributions

1. **Expository:** A fully rigorous, self-contained proof of gradient learning convergence to Nash in general multi-state games (Chapters 2–3)
2. **Original:** Global convergence via restarts (Theorem 4.2) — removes dependence on initialisation
3. **Empirical:** Validation on normal-form and stochastic games, demonstrating:
   - Basin probability $p \approx 1$ — one restart almost always suffices in practice
   - Best-of-K restarts reliably selects the Pareto-optimal (best joint) equilibrium
   - Ignoring state structure costs 38% of achievable social welfare in the 2-state game

Where Next?

The most compelling open direction:

Meta-learning — agents that don't just respond to opponents, but *anticipate* how opponents will learn and actively steer their updates.

This is the mechanism behind **algorithmic collusion**: an agent that teaches its competitor to cooperate by shaping how the competitor's gradient updates unfold.

This dissertation provides the convergence foundation that makes such analysis rigorous.

Other directions:

- Neural network policies
- Sample complexity bounds
- Smarter restart strategies
- Continuous action spaces

Local convergence + random restarts = global convergence.
Simple, provably correct, empirically effective.

Thank You

Thank you for your attention.

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